ETHAN JANTZ

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EXPERIENCE

Senior Data Engineer Consultant | Skeleton Key Strategies
Jan 2026 – Present

- Sole engineer on a bidirectional data synchronization service between Salesforce and NGPVAN, handling contacts, events, and registrations across ~8M records on a 3-hour sync cycle for canvassing and outreach operations.
- Designed and executed a migration of ~8M constituent records from Salesforce to NGPVAN, including data mapping, validation, and deduplication logic.
- Drove requirements definition and business logic design with non-technical stakeholders for complex merge, conflict resolution, and bidirectional sync rules across two CRM systems.
- Built a comprehensive testing and observability framework — including audit logging, run-level diagnostics, and staged output validation — to enable production-safe iteration in an environment with no dedicated testing infrastructure.
- Served as technical liaison between client and external vendor, coordinating requirements, translating technical constraints for non-technical stakeholders, and managing delivery timelines.
- Produced architectural diagrams and system documentation communicating data flow, sync logic, and error handling to both technical and non-technical audiences.

Recurser | The Recurse Center
Jul 2025 – Dec 2025

- Built and deployed Deed Dish (deeddish.com), an interactive GIS application for exploring property ownership in Chicago. Designed a spatial ETL pipeline in Python to scrape, geocode, validate, and process ~7M parcel records from Cook County, with DuckDB for spatial queries, Cloudflare R2 for document storage, and a TypeScript/MapLibre GL frontend enabling map-based exploration of ownership patterns across geographic layers.
- Developed a semantic word game solver using Word2Vec embeddings and cosine similarity that achieved 100% accuracy within 3 guesses, significantly outperforming human players.
- Contributed to an open-source AI social deduction game, refactoring direct OpenAI API calls to use LangChain, enabling easy experimentation across multiple LLM providers.
- Founded and led a weekly code golf group (5–10 attendees) focused on deepening Python fluency through collaborative problem-solving.

U.S. Digital Corps Data Science and Analytics Fellow | Office of Head Start
Jul 2023 – Jul 2025

- Informed $805M in grant funding allocation through statistical and geospatial analysis, including spatial modeling of service area coverage and demographic need, supporting critical leadership decisions and program oversight.
- Conducted GIS-based feasibility analysis assessing the viability of utilizing Head Start centers near military bases to provide childcare for families living on base, integrating facility location data with geographic proximity and demographic datasets.
- Led design and deployment of ETL pipelines for the Head Start Enterprise System on AWS, architecting scalable cloud-based data infrastructure using Python and implementing comprehensive testing with pytest.
- Oversaw contractor development of data warehouse infrastructure, defining requirements, coordinating technical delivery, and managing vendor relationships to support the organization’s analytics platform.
- Built automated data processing pipelines using SQL (MySQL, PostgreSQL), Tableau, Python, R, and Quarto Markdown, implementing data validation and quality protocols that streamlined reporting workflows.
- Developed a full-stack dashboard application, collaborating with frontend developers and non-technical stakeholders to deliver real-time program monitoring capabilities.

Associate Research Analyst | Chicago Metropolitan Agency for Planning

Jul 2022 – Jul 2023

- Led migration of 150 spatial datasets from the organization's open data portal to ArcGIS Hub, managing vendor coordination and developing data governance policies, data dictionaries, and metadata standards — introducing formal data stewardship practices to the organization.
- Automated geospatial data processing workflows using Python and SQL while architecting Azure cloud storage solutions, reducing manual reporting efforts and improving spatial data infrastructure scalability.
- Performed spatial and demographic analysis using R and Python to support regional planning initiatives, including freight route impact analysis on Chicago's South Side and annual Census-based demographic updates integrating public and partner datasets.
- Provided technical assistance to planning staff using the organization's internal R-based mapping and spatial visualization tools, supporting their analytical workflows.
- Delivered technical training to 20 staff across two departments on R and git, building internal data analysis and GIS capacity.

Research Assistant | Nathalie P. Voorhees Center

Sep 2019 – Jul 2022

- Engineered an interactive web-based GIS dashboard using R Shiny, CSS, and JavaScript to visualize block-level disability accessibility metrics across 200 cities, mapping Census-derived indicators of where people with disabilities live against transportation accessibility gaps.
- Designed and implemented spatial ETL workflows for transportation and Census data with R and Bash, enabling reproducible geospatial data processing pipelines.
- Built research databases supporting transportation, demographic, and housing spatial research using PostgreSQL (with PostGIS) and MongoDB.
- Visualized data for the Illinois Interagency Task Force on Homelessness's 2023 annual report.

EDUCATION

University of Illinois at Chicago

Master of Science in Civic Analytics | 2022

Bachelor of Arts in Urban Studies | 2020

SKILLS

GIS (ArcGIS, QGIS, GDAL, PostGIS, GeoPandas, Shapely, geocoding, spatial joins), Python, C/C++, Javascript/Typescript, HTML/CSS, R, SQL (Postgres, MySQL, RedShift, DBT), Docker, AWS, Azure, Salesforce, Civis Analytics, NGPVAN EveryAction, data governance, technical documentation